Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 1 of 7 · Question 3 part c · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number				
3	(a) (i)	water;		1
	(ii)	• allows urea to pass; • prevents other larger molecules from passing;	Allow small molecules go through	2
	(iii)	• breaks down urea; • to produce ammonium ions; • to react with/stimulate sensor;	Allow ammonia Allow idea of being picked up/detected by sensor	3
	(b)	• enzyme activity increases/enzyme works faster/higher rate of reaction; • at higher temperatures; • more ammonium ions produced; • higher reading;	Ignore details related to enzyme denaturing / collision theory/kinetic energy ORA for lower temperature	4
	(c)	any two from • can be reused; • products not contaminated; • enzymes more stable; • cost effective justified;	Allow can withstand temperature/pH changes	2

Total 12 marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 1 of 7 · Question 3 part c · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 2 of 7 · Question 6 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

6 (a)	(i)	sucrase/invertase (1)		1
	(ii)	• enzyme is a protein (1) • biuret test (1) • purple if protein present/remains blue if absent (1)		3
	(iii)	• reusable (1) • stable (1)		2
(b)	(i)	• allows glucose molecules to pass (1) • prevents larger/other molecules/ blood cells from passing (1)		2
	(ii)	• breaks down/converts glucose (1) • into hydrogen peroxide (1) • which activates electrode (1) • more glucose results in more hydrogen peroxide (1)		4

Total 12 marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 2 of 7 · Question 6 part a.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 3 of 7 · Question 6 part a.iii · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

6 (a)	(i)	sucrase/invertase (1)		1
	(ii)	• enzyme is a protein (1) • biuret test (1) • purple if protein present/remains blue if absent (1)		3
	(iii)	• reusable (1) • stable (1)		2
(b)	(i)	• allows glucose molecules to pass (1) • prevents larger/other molecules/ blood cells from passing (1)		2
	(ii)	• breaks down/converts glucose (1) • into hydrogen peroxide (1) • which activates electrode (1) • more glucose results in more hydrogen peroxide (1)		4

Total 12 marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 3 of 7 · Question 6 part a.iii · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 4 of 7 · Question 7 part a.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 4 of 7 · Question 7 part a.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 4 of 7 · Question 7 part a.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 5 of 7 · Question 7 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 5 of 7 · Question 7 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 5 of 7 · Question 7 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 6 of 7 · Question 6 part c.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

Number				
6	(a) (i)	18-12; 6;	accept 17.8/17.9	2
	(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
	(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
	(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)	Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c) (i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
	(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 6 of 7 · Question 6 part c.i · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 7 of 7 · Question 7 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 7 of 7 · Question 7 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Make alginate beads and give uses and advantages of immobilised enzymes

Answer 7 of 7 · Question 7 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks